

System Design

A complete, implementable specification of the improved conversational-memory system — every component tied to a capability, invariant, or measured failure.

System Conversational Memory Intelligence System

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Envelope small/team B2B SaaS (ADR-004)

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api_contracts.md · threat_model.md

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DESIGN IN ONE PARAGRAPH

A write path **extracts typed, resolved facts** from conversation, screens them for PII at admission, and stores them with **validity intervals and provenance** in a single transactional store. A read path retrieves **tenant-isolated, currently-valid** facts, ranks them on four hand-specified signals, and constructs a **budgeted, position-aware** injection — or abstains. Conflict resolution lives on the **write path** (invalidation), not the ranker. Isolation is a **row-level invariant**, not a filter. Every stage emits a trace, and the whole system is measured offline against the Deliverable 3 baseline. Each choice below cites the capability (C1–C10), premise (P1–P8), or measured failure it answers.

1. Inputs, outputs, actors, boundaries, trust

INPUTS

Conversation turns (utterances) scoped to a tenant, account, and user; explicit user corrections; deletion requests. **P3** Input is low-precision, high-redundancy, and may contain PII or contradictions — it is **untrusted**.

OUTPUTS

A budgeted, ordered set of memories to inject into the agent's context (or an abstention); admission decisions; a structured decision trace; deletion confirmations with a measured window.

ACTORS

The **GTM agent** (calls admit on write, retrieve on read); the **end user** (AE / SDR / RevOps, per tenant); the **data-protection officer / admin** (deletion, audit); the **vendor operator** (runs the service). No actor can name another tenant.

BOUNDARIES & TRUST

TB-1 admission: nothing crosses into storage until it is typed (C3) and PII-screened (C2/F11). **TB-2 isolation:** no code path can express a cross-tenant read (C8/F10) — enforced by RLS + repository scoping, not by callers remembering to filter. Raw transcripts live outside this system; we hold only *observed* facts + provenance references.

2. Memory admission policy and representation

Admission (C2, fixes F8/F11). "Store everything" is a policy, and it is the one that maximises noise (P3). Every candidate passes three gates before storage: **extract** (utterance → typed resolved claim, fixing F7 — a chunk is not a fact), **PII screen** (Presidio deterministic- first, a hard gate — post-hoc redaction is impossible

once data is indexed, ADR-003), and **op-select** (store / update / discard vs existing facts, deduping — Mem0-style, prototype).

Representation (C3, fixes F4/F5/F7). A memory is not a proposition; it is a proposition + a validity interval + a supersession relation (P4). Each fact carries type, subject, content, confidence, observation time, bi-temporal validity, provenance, and tenant — full schema in `data_model.md`. Provenance is what makes consolidation safe and deletion complete.

3. Storage and index

Single Postgres + pgvector (ADR-005). The `memory` table is the source of truth; `memory_embedding` is a *derived, rebuildable* pgvector index (C4). At the small/team scale ($\sim 2 \times 10^6$ facts, ADR-004) both live in one transactional database, which collapses the deletion consistency window (F9) to a single transaction and lets isolation be a row-level invariant (F10). An embedding-model change is a re-projection, not a migration.

TRADE This is tier-bound: mid-market forces a dedicated vector store and reopens the deletion-window problem (ADR-005 revisit).

4. Ranking signals and conflict resolution

Conflict resolution is on the write path, not the ranker (ADR-001). A contradicting fact *invalidates* the prior one at admission (sets its `valid_to`), so retrieval only ever sees currently-valid facts. This is the direct fix for the D3 result that a stale fact beat the current one **80%** of the time — similarity cannot encode validity (F4), so validity is removed from the ranker's job.

Multi-signal ranking (C5). The ranker orders already-valid facts on four **hand-specified, inspectable** signals (D1 non-goal: no learned ranker in v1): **relevance** (vector similarity), **recency** (observation time), **importance** (by type — `requirement` highest, so the buried EU-residency dealbreaker wins its slot), and **confidence** (corrections = 1.0). Weights are a config artifact, visible in every trace (C9).

5. Context selection and token budgeting

Budgeted construction (C6, fixes F2/F3). Injection is a constrained selection problem, not a copy (P2). Memories are added in rank order until the **≤ 2000 -token** budget binds; ordering is **position-aware** (highest-value at the ends — the middle of a long context is where models use facts least, per *Lost in the Middle*).

Defined edge behaviour: when nothing clears the **abstention floor**, return an explicit abstention (the D3 cold-start fix — the baseline failed this 2/2, injecting a foreign tenant's budget and a PII record on a query with nothing on file).

6. Lifecycle: update, consolidate, reflect, expire, delete

OPERATION	FIXES	BEHAVIOUR
Update / correct	F1	Supersession transaction; a user correction is the highest-confidence signal there is (C1) — if nothing else is written, corrections are.
Consolidate	F5, F6	Cite, don't replace: a rollup memory links to its sources, which stay retrievable. This is the structural guard against summarisation loss and drift (C7). Highest-risk component (first_principles §5) → earliest adversarial testing.
Reflect	—	Deliberately minimal in v1 (D1 §5.3: validity does more work than reflection). Deferred.
Expire	—	Validity-end and commitment-due sweeps set <code>valid_to</code> ; rows kept for audit.
Delete	F9	Hard delete cascades source + index in one transaction; the window is measured per request, not assumed (ADR-004/005).

7. PII, authorization, tenant isolation, auditability

PII (C2/F11, ADR-003). Deterministic recognizers first (no LLM on the PII path, no injection surface); a hard admission gate, with deletion as backstop. **Isolation (C8/F10, ADR-005).** `tenant_id` is enforced twice: Postgres **Row-Level Security** (a raw query that forgets the predicate still reads nothing foreign) and a **repository whose tenant is a constructor argument, never a method parameter** — so no method signature in the codebase can express a cross-tenant read. **D3 EVIDENCE** The naive baseline leaked foreign-tenant memories on 7/11 queries; a stronger embedder makes that *worse*, so isolation must be structural. **Audit (C9).** Every read writes an `access_log` row: who saw what, when.

8. Observability and offline evaluation

Observability (C9, from P7). The system's decisions are invisible in its output, so every stage emits a structured event: candidates considered, per-signal scores, budget drops, PII suppressions, final set — retrievable via `/trace/{id}`. This is what lets the D3 failure taxonomy be populated from real traffic. **Offline eval (C10, from P8, ADR-002).** A fixed, versioned adversarial set (the D3 dataset + LongMemEval scaffold) runs the **3-arm** comparison (naive / +recency / validity-filter) against the baseline — the non-negotiable gate. **NOTE** C9 and C10 are the two capabilities the failure analysis could *not* surface (first_principles §5) — they exist to make everything else debuggable and provable.

9. Scale, latency, cost, failure recovery

DIMENSION	V1 TARGET (ADR-004)	DESIGN CONSEQUENCE
Corpus	$\leq \sim 2 \times 10^6$ typed facts	single Postgres + pgvector HNSW (ADR-005)
Injection budget	≤ 2000 of 16000 tokens	budgeted, position-aware construction (C6)
Latency	p95 ≤ 300 ms retrieve+rank+construct	ANN + in-DB filter; no per-candidate LLM on the read path
Cost	$\leq 15\%$ of per-turn token spend	LLM used only at admission (extract, op-select), bounded & batched
Deletion	near-synchronous; ≤ 24 h backstop	same-txn cascade (F9); window measured
Cross-tenant leak	zero, tested	RLS + repo scoping (C8)

Failure recovery. Extractor/embedder outage → admission **queues** candidates (a correction must never be lost, C1), reads continue on existing facts. Index loss → rebuild from source of truth (C4). Bad invalidation (the ADR-001 risk) → history is preserved, so an incorrectly superseded fact is recoverable by re-validation, not gone.

10. API contracts & data migrations

Full contracts in `api_contracts.md`: `/memory/admit`, `/memory/retrieve`, lifecycle jobs, `/deletion`, `/trace/{id}`, `/eval/run`. The load-bearing contract rule: **tenant is ambient, never a request field** — there is no wire to spoof. Migrations: schema changes are ordinary Postgres migrations; an embedding-model swap is a re-projection of `memory_embedding` (not a data migration), per C4.

11. Milestones with acceptance tests

M	MILESTONE	ACCEPTANCE TEST (GATE)
M1	Schema + isolation skeleton (<code>data_model</code> , RLS, repository)	adversarial cross-tenant suite returns zero foreign rows via API, raw-omitted-predicate, and mis-scoped repo (F10)
M2	Write path: <code>extract</code> → PII gate → <code>op-select</code> → store	PII candidate stores nothing; supersession txn invalidates prior fact (F11, F4)
M3	Read path: <code>retrieve</code> → rank → budget → abstain	beats the D3 baseline on the fixed set; abstains on cold-start; <code>used_tokens</code> ≤ budget (C5/C6/C10)
M4	Lifecycle: consolidate (cited), expire, delete	deletion window within guarantee; consolidation keeps sources retrievable (F9, F5/F6)
M5	Observability + eval harness (3-arm)	a trace localises an injected failure in D6; 3-arm report reproduces (C9/C10)

COMPLETION STANDARD (HANDBOOK §6.2)

Another engineer can implement this without unstated verbal context: each component has a justified purpose (the C/P/F citation), interfaces are in `api_contracts.md`, the schema in `data_model.md`, security boundaries and the threat model in `threat_model.md`, and every open call is an ADR. Where a decision is still provisional (op-selector cost, consolidation) it is marked *prototype* in the backlog with a validation plan, not hidden.

AI assistance disclosure. Claude helped draft this specification. The architecture, the component→capability→failure mapping, the isolation and lifecycle decisions, and the scope choices (ADR-004/005) are mine and are what I will defend in review. Every component traces to a capability, premise, or measured failure from Deliverables 1 and 3.